

IntentLink API

1.0.0 OAS 3.1

</api/openapi.json>

API for parsing, planning, and executing DeFi intents securely.

v1**POST****/api/v1/parse-intent/** Parse a Natural Language Intent

Parses a natural language input into a structured Intent object.

Phase 1 Implementation: This is a stubbed endpoint. It does not call an LLM. It returns a hardcoded, deterministic response for a specific known input. This allows frontend development to proceed.

Parameters**Cancel****Reset**

No parameters

Request body required**application/json****Edit Value** Schema

```
{
  "input": "Stake 1000 BDAG in the safest farm",
  "user_wallet": "0xAb5801a7D398351b8bE11C439e05C5B3259aeC9B",
  "chain_id": 1043
}
```





Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8001/api/v1/parse-intent/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "input": "Stake 1000 BDAG in the safest farm",
    "user_wallet": "0xAb5801a7D398351b8bE11C439e05C5B3259aeC9B",
    "chain_id": 1043
  }'
```

Request URL

http://localhost:8001/api/v1/parse-intent/

Server response

Code	Details
200	Response body <pre>{ "intent_id": "a3235822-f35d-4c24-b722-bcc79458afff", "status": "parsed", "intent": { "intent_type": "stake_and_compound", "asset": "BDAG", "amount": 1000, "amount_unit": "token", "target": "highest_risk_adjusted_apr" }, "clarify_questions": [] }</pre> Download Response headers <pre>connection: close content-length: 245 content-type: application/json; charset=utf-8 cross-origin-opener-policy: same-origin date: Wed, 19 Nov 2025 11:26:20 GMT referrer-policy: same-origin server: gunicorn x-content-type-options: nosniff x-frame-options: DENY</pre>

Responses

Code	Description	Links
200	OK	No links
Media type		



Code

Description

Links

application/json

Controls Accept header.

Example Value Schema

```
{
  "intent_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "status": "string",
  "intent": {
    "intent_type": "string",
    "asset": "string",
    "amount": 0,
    "amount_unit": "string",
    "target": "string"
  },
  "clarify_questions": [
    "string"
  ]
}
```

POST**/api/v1/plan/** Generate an Execution Plan

Takes an intent_id and generates a ranked list of candidate execution plans using GoPlus for security validation.

Phase 1 Implementation (updated):

- Uses an explicit whitelist of protocol contracts (staking + lending) provided by the blockchain team for BlockDAG Awakening Testnet.
- Calls SecurityService (GoPlus) for safety_score + warnings.
- Uses mock APY/TVL values for ranking/utility computation.
- Supports selecting either staking or lending candidates based on parsed intent.

Parameters**Cancel****Reset**

No parameters

Request body **required****application/json****Edit Value** Schema



```
{
  "intent_id": "a3235822-f35d-4c24-b722-bcc79458afff"
}
```

**Execute****Clear**

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8001/api/v1/plan/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "intent_id": "a3235822-f35d-4c24-b722-bcc79458afff"
  }'
```

Request URL

http://localhost:8001/api/v1/plan/

Server response

Code

Details

200

Response body

```
{
  "plan_id": "f25e5c8b-d491-444f-b1e4-5cbe9c19d67f",
  "candidates": [
    {
      "address": "0x1b227DF9c8D34CaB880774737FBf426E66Ba98Ed",
      "apy": 0.12,
      "tvl": 500000,
      "safety_score": 90,
      "utility": 0.51,
      "warnings": [
        "Could not retrieve a token security report from GoPlus."
      ],
      "protocol": "staking"
    }
  ]
}
```



Code

Details

```
}
],
"chosen": {
  "address": "0x1b227DF9c8D34CaB880774737FBf426E66Ba98Ed",
  "apy": 0.12,
  "tv1": 500000,
  "safety_score": 90,
  "utility": 0.51,
  "warnings": [
    "Could not retrieve a token security report from GoPlus."
  ],
  "protocol": "staking"
}
}
```

Download

Response headers

```
connection: close
content-length: 521
content-type: application/json; charset=utf-8
cross-origin-opener-policy: same-origin
date: Wed, 19 Nov 2025 11:27:41 GMT
referrer-policy: same-origin
server: gunicorn
x-content-type-options: nosniff
x-frame-options: DENY
```

Responses

Code	Description	Links
200	OK	No links

Media type

application/json

Controls Accept header.

Example Value Schema

```
{
  "plan_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "candidates": [
    {
      "address": "string",
      "apy": 0,
      "tv1": 0,
      "safety_score": 0,
      "utility": 0,
      "warnings": [
        "string"
      ],
      "protocol": "string"
    }
  ],
  "chosen": {
    "address": "string",
    "apy": 0,
    "tv1": 0,
    "safety_score": 0,
    "utility": 0,
    "warnings": [
      "string"
    ]
  }
}
```



Code

Description

Links

```
    "protocol": "string"
  }
}
```

POST

/api/v1/submit-intent/ Submit a Plan for Execution



Accepts a plan_id, creates an Execution record, and queues the async task.

Parameters

Cancel

Reset

No parameters

Request body **required**

application/json

Edit Value Schema

```
{
  "plan_id": "f25e5c8b-d491-444f-b1e4-5cbe9c19d67f"
}
```



Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8001/api/v1/submit-intent/' \
```



```
-H 'accept: application/json' \  
-H 'Content-Type: application/json' \  
-d '{  
  "plan_id": "f25e5c8b-d491-444f-b1e4-5cbe9c19d67f"  
}',
```

Request URL

http://localhost:8001/api/v1/submit-intent/

Server response

Code	Details
200	Response body <pre>{ "execution_id": "01b63dd8-911e-4950-a98c-0424e8ea68e4", "status": "pending" }</pre> Download Response headers <pre>connection: close content-length: 77 content-type: application/json; charset=utf-8 cross-origin-opener-policy: same-origin date: Wed,19 Nov 2025 11:30:12 GMT referrer-policy: same-origin server: gunicorn x-content-type-options: nosniff x-frame-options: DENY</pre>

Responses

Code	Description	Links
200	OK Media type application/json Controls Accept header. Example Value Schema <pre>{ "execution_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6", "status": "string" }</pre>	No links

GET /api/v1/execution/{execution_id}/status/ Poll Execution Status



Returns the current status and tx_hash of an execution.

Parameters

Cancel



Name	Description
execution_id * required string (path)	01b63dd8-911e-4950-a98c-0424e8ea68e4

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8001/api/v1/execution/01b63dd8-911e-4950-a98c-0424e8ea68e4/status/' \
  -H 'accept: application/json'
```

Request URL

http://localhost:8001/api/v1/execution/01b63dd8-911e-4950-a98c-0424e8ea68e4/status/

Server response

Code	Details
200	Response body <pre>{ "execution_id": "01b63dd8-911e-4950-a98c-0424e8ea68e4", "status": "confirmed", "tx_hash": "0x01b63dd8911e4950a98c0424e8ea68e4123456789", "logs": ["Transfer", "Stake", "IntentExecuted"] }</pre> Download Response headers <pre>connection: close content-length: 186 content-type: application/json; charset=utf-8 cross-origin-opener-policy: same-origin date: Wed, 19 Nov 2025 11:31:48 GMT referrer-policy: same-origin server: unicorn x-content-type-options: nosniff x-frame-options: DENY</pre>

Responses



Code	Description	Links
200	OK Media type application/json Controls Accept header. Example Value Schema <pre>{ "execution_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6", "status": "string", "tx_hash": "string", "logs": []}</pre>	No links

Schemas



- IntentParseOutput > Expand all object
- IntentSchema > Expand all object
- IntentParseInput > Expand all object
- CandidateSchema > Expand all object
- PlanOutput > Expand all object
- PlanInput > Expand all object
- SubmitIntentOutput > Expand all object
- SubmitIntentInput > Expand all object
- ExecutionStatusOutput > Expand all object